

## HOWTO — AIStudio User Guide

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Practical answers for day-to-day AIStudio use. Not a getting-started guide (see [QUICKSTART.md](#)) — reach for this when you need to do something specific or something isn't working as expected.

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### Shell & Terminal

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**What is the shell?** The shell (Terminal) is a text-based interface for running commands on your Mac. AIStudio setup and management uses the shell for tasks the UI doesn't cover — like installing models or ingesting a new corpus. New to the shell? See [Apple's Terminal User Guide](#).

**How do I open the Terminal?** Press `Cmd+Space`, type `Terminal`, press Enter. Or open Finder → Applications → Utilities → Terminal.

**Why does my terminal say "command not found" when I paste commands?** Two common causes: - You pasted a `#` comment line — zsh tries to execute it. Paste only the commands, not the comment lines. - Your AIStudio aliases aren't loaded yet. Run:

```
source ~/.zshrc
```

Then try the command again on its own line. Do not chain: `source ~/.zshrc && ais_start` will fail. Run `source ~/.zshrc` first, then `ais_start`.

## Using AIStudio

After running `./ais_install` from the repo root (see QUICKSTART.md), these commands are available from any terminal. If a command is not found, run `source ~/.zshrc` first.

| Command                           | What it does                                                                                                                                  |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ais_install [cmd]</code>    | Install AIStudio user commands — <code>ais_install ais_log</code> adds a single command, <code>ais_install --verify</code> checks all aliases |
| <code>ais_start</code>            | Start all services and open the UI in your browser                                                                                            |
| <code>ais_stop</code>             | Stop all services                                                                                                                             |
| <code>ais_bench</code>            | Run a benchmark on the demo corpus                                                                                                            |
| <code>ais_log</code>              | Tail live AIStudio backend log — run in a separate tab after <code>ais_start</code>                                                           |
| <code>ais_download_sec_10k</code> | Download SEC 10-K filings from EDGAR to <code>~/Downloads/sec_10k/</code> (~2 GB)                                                             |
| <code>ais_help</code>             | Print this command reference                                                                                                                  |

Every command supports `--help`:

```
ais_start --help
ais_bench --help
```

## Upgrading AIStudio

**How do I upgrade AIStudio when a new version is released?**

Pull the latest code from GitHub:

```
cd ~/Developer/AIStudio && git pull
```

Then update Python dependencies in case new packages were added:

```
source .venv/bin/activate && pip install -r requirements.txt
```

Then restart:

```
ais_start
```

**Does upgrading delete my corpus data?** No. Your corpus data lives in `~/qdrant_storage/` on disk — completely separate from the AIStudio codebase. It persists across upgrades, restarts, and reboots. You never need to re-ingest after a routine upgrade.

**When would I need to re-ingest after an upgrade?** Only if the release notes say the chunk format changed. This is rare and always announced. When it happens, re-ingest from the UI: open the corpus, **Delete Corpus**, recreate it with the same name, and re-upload your files with **Add** — AIStudio rebuilds it cleanly in one pass. The `--force` flag wipes and rebuilds the corpus cleanly.

**The demo corpus re-ingests automatically** on first `ais_start` after an upgrade if needed.

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## Corpus Management

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**How do I ingest a new corpus?** Use the UI — open AIStudio, create a new corpus using the **New** button, then upload your files using the **Add** button. AIStudio handles ingestion automatically and shows progress in the chat area.

**How do I re-ingest a corpus from scratch?** Delete the corpus using the **Delete Corpus** button in the UI (type YES to confirm — it moves to Trash, recoverable). Then create a new corpus with the same name and re-upload your files via **Add**. AIStudio ingests everything in one pass.

**How do I delete a file from a corpus?** Use the UI — select your corpus, click **Inspect** to open the file list, then click **Delete** next to the file you want to remove. AIStudio moves the file to trash and removes its chunks from the vector index. You do not need to re-ingest the rest of the corpus — only the deleted file's chunks are removed.

**What happens when I delete a file from a corpus — is it gone forever?** No. Deleted files move to `data/corpora/<name>/uploads/trash/` — not permanently deleted.

**How do I recover a file I accidentally deleted from a corpus?**

```
# See what's in trash
ls ~/Developer/AIStudio/data/corpora/<name>/uploads/trash/

# Move it back
```

```
mv ~/Developer/AIStudio/data/corpora/<name>/uploads/trash/<filename> \  
    ~/Developer/AIStudio/data/corpora/<name>/uploads/  
  
# Then re-upload the file via the UI Add button to restore it
```

**What happens when I delete an entire corpus — is it gone forever?** No. The corpus folder moves to `~/Trash/`. The folder is renamed to avoid conflicts with previous deletions — it will be named `AIStudio_<name>` or `AIStudio_<name>_<timestamp>` if a folder with that name already exists in Trash.

**How do I recover a corpus I accidentally deleted?** First find the folder in Trash — it may have a timestamp suffix:

```
ls ~/.Trash/ | grep AIStudio_<name>
```

Then move it back and re-ingest:

```
mv ~/.Trash/AIStudio_<name> ~/Developer/AIStudio/data/corpora/<name>  
# Or if it has a timestamp suffix:  
mv ~/.Trash/AIStudio_<name>_<timestamp> ~/Developer/AIStudio/data/corpora/<name>  
# Then re-ingest
```

Then re-upload the files via the UI (Add button) to restore the corpus in Qdrant.

### How do I ingest the SEC 10-K corpus?

The SEC 10-K corpus is a special case. AIStudio provides `ais_download_sec_10k` because the SEC EDGAR filing system uses a specific access protocol — this automates what would otherwise be a complex multi-step download. But the ingest step that follows is **identical to ingesting any corpus you bring yourself**. The download step is what's special.

This is one of the few places in AIStudio where you'll run terminal commands directly. We expose it deliberately — understanding how large corpora are built is part of what makes AIStudio a learning tool, not just a product.

#### Step 1 — Download the filings to `~/Downloads/sec_10k/` (~5 min, ~2 GB):

```
ais_download_sec_10k
```

#### Step 2 — Ingest using the AIStudio UI:

Open AIStudio, create a new corpus named `sec_10k`, then upload the files from `~/Downloads/sec_10k/` using the **Upload** button. This is the same process as ingesting any corpus you build yourself — the download step is what's special.

Allow ~30 minutes for ingestion to complete.

**Why ~/Downloads?** You own the downloaded files — ~2 GB of SEC filings you may want to reuse, inspect, or build a different subset from. They stay in `~/Downloads/sec_10k/` until you decide what to do with them.

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## Installing and Managing LLMs

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*How do I see what models are installed?*

```
ollama list
```

*How do I install a new LLM?*

```
ollama pull gemma3:4b           # ~3.3GB — the out-of-the-box default, fast
ollama pull gemma3:27b          # ~17GB — best quality-for-size; the benchmark model
ollama pull llama3.1:8b         # ~5GB — alternative, fast
ollama pull llama3.1:70b        # ~42GB — best quality, requires ~64GB RAM
ollama pull mistral:7b          # ~4.4GB — alternative option
```

Once pulled, the model appears automatically in the **Model** dropdown in the UI. No restart required.

**Which model should I choose?** On Apple Silicon, `llama3.1:70b` and `llama3.1:8b` have identical query latency (~6–7s) once warm. Choose based on available RAM: - `gemma3:4b` — 8GB+ RAM, the out-of-the-box default; small, fast, best first run - `gemma3:27b` — 32GB+ RAM, best quality-for-size; the model AIStudio benchmarks on - `llama3.1:8b` — 8GB RAM minimum, solid alternative - `llama3.1:70b` — 64GB+ RAM, best answer quality - `mistral:7b` — good alternative on constrained hardware

*How do I remove a model I no longer need?*

```
ollama rm mistral:7b
```

See the [Ollama model library](#) for all available models.

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## Query Settings

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The **Query Settings** panel in the sidebar controls how AIStudio retrieves and generates answers. These settings apply to every query until you change them.

**Top K** — *How many document chunks to retrieve* The number of passages retrieved from your corpus before generating an answer. Higher values give the model more context but increase latency slightly. - Default: 5 — good for most queries - Try 10 for complex questions spanning multiple documents - Try 3 for faster responses on focused questions

**Choosing Top K by query type:**

| Query type                  | Recommended Top K | Example                                                   |
|-----------------------------|-------------------|-----------------------------------------------------------|
| Point-in-time, single firm  | 5                 | "What was Goldman's CET1 ratio in 2024?"                  |
| Trend across multiple years | 10–15             | "How has JPMorgan's capital ratio changed 2022–2025?"     |
| Cross-firm comparison       | 10                | "Compare risk disclosures at JPMorgan, Goldman, and Citi" |
| General question            | 5                 | "What is the company's AI strategy?"                      |

If an answer seems incomplete or missing years/firms you expected — increase Top K first before rephrasing the question.

Adjust using the **Top K** input field in the sidebar.

**Temperature** — *How creative vs. precise the answer is* Controls randomness in the model's output. - 0.1–0.3 — precise, factual, stays close to the source documents. Best for financial data, specific numbers, compliance questions. - 0.5–0.7 — more varied, synthesizes across sources. Better for summaries. - Default: 0.3 — recommended starting point Adjust using the **Temperature** input field in the sidebar.

**Retrieval Mix** — *Balance between literal and conceptual search* Controls how the system finds relevant passages in your corpus.

- **Literal** (slider left, value 0.0) — prioritizes exact word and phrase matching (BM25). Use when you know the specific term, entity name, or abbreviation that appears in your documents. Example: searching for "CET1" by exact term.
- **Conceptual** (slider right, value 1.0) — prioritizes semantic meaning. Finds passages related to your query even when the document uses different phrasing. Example: asking about "capital strength" returns passages about CET1, Tier 1 capital, and capital ratios.
- **Default 0.5** — blends both. Works well for most queries.

**Choosing by query type:**

| Query type                       | Recommended setting  | Why                                                    |
|----------------------------------|----------------------|--------------------------------------------------------|
| Named entity, ticker, exact term | Literal (0.0–0.3)    | BM25 matches the token directly                        |
| Thematic or conceptual question  | Conceptual (0.7–1.0) | Semantic search handles paraphrase                     |
| Multi-firm comparison            | Center (0.4–0.6)     | Entity names + topic meaning both matter               |
| Single firm, multiple years      | Conceptual (0.8–1.0) | Year context is in documents; meaning guides retrieval |

Adjust using the **Retrieval Mix** slider in the Query Settings sidebar.

**Score Threshold** — *Relevance floor (per-corpus default)* Chunks scoring below this are dropped before the model ever sees them. Default `0.50`. Lower it ( $\approx 0.2-0.3$ ) if good answers come back empty or hedged; raise it to cut noise. Unlike the three settings above, Score Threshold is a **per-corpus default**, not a per-query control — set it in **Edit** → **Query Defaults** for the selected corpus, where it applies whenever that corpus is queried.

## Understanding Citations

Every answer includes numbered citations — `[1]`, `[2]`, `[3]` — in the text. These refer to the source passages listed in the **References** panel on the right side of the screen.

**What citations tell you:** - Which document the information came from - Which page in that document

**What to do when an answer seems wrong:** Check the citations first. If the referenced chunks don't actually contain the claim — the model may be reasoning beyond its sources. Try increasing Top K, adjusting the Retrieval Mix toward Literal, or rephrasing to be more specific.

**Uncited claims:** if a sentence has no citation marker, the model is drawing on general context rather than a specific retrieved passage. Treat uncited claims with more caution, especially for specific numbers or facts.

**Opening the source:** click the citation chip in the References panel to see the full chunk text and open the source document.

# Benchmark & Corpus Testing

The benchmark harness serves two purposes: - **Performance** — measures query latency per question - **Veracity** — validates answer quality against a questions file

## How do I run a benchmark on the demo corpus?

```
ais_bench
```

Parameters (Top K, Temperature, Retrieval Mix, Score Threshold) are read automatically from corpus metadata — no flags needed for built-in corpora.

## How do I benchmark a different corpus?

```
ais_bench --corpus sec_10k
ais_bench --corpus help
```

## How do I test with a specific model or override settings?

```
ais_bench --corpus demo --model llama3.1:70b
ais_bench --corpus demo --alpha 0.3 --min-score 0.2
ais_bench --help # all flags
```

**What makes a benchmark question pass or fail?** Three checks — all must pass: 1. All `keywords` appear in the answer (case-insensitive) 2. The answer includes at least one citation 3. The model doesn't hedge with "no information available" or similar phrases

**How do I write my own test questions for a corpus?** Create `benchmarks/<corpus_name>/<corpus_name>_questions.yaml`:

```
- topic: Your Topic
  questions:
    - id: unique_snake_case_id
      question: The exact question text sent to AISTudio.
      keywords: [term1, term2, term3] # all must appear in the answer to pass
      notes: What a correct answer looks like — which document, what content.
```

Run `ais_bench --corpus <name>` — the questions file is auto-detected by corpus name.

**Tips for good keywords:** use 2–5 distinctive terms that prove the model retrieved the right content — specific concepts, entity names, or regulatory terms. Avoid generic words that would appear in any answer.

**Where do reports go?** `benchmarks/<corpus>/reports/` — timestamped `.md`, `.json`, and `.pdf` files. The `.md` shows pass/fail per question with latency, citations, and the answer text.



### PDF reports aren't being generated — I see "PDF skipped"

`weasyprint` is required for PDF output and must be installed separately:

```
pip3 install weasyprint --break-system-packages
```

It's already in `requirements.txt` but not installed by default. Once installed, subsequent runs generate `.pdf` reports alongside `.md` and `.json`. The `PDF skipped` message is printed when `weasyprint` is missing — `.md` and `.json` reports are always generated regardless.

See `TUTORIAL.md §3.4` for a step-by-step walkthrough.

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## Troubleshooting

### UI shows "Error loading corpora" or "Ollama not running" on startup

The browser opened before the backend finished starting. Hard-refresh ( `Cmd+Shift+R` ). If that doesn't fix it, check the terminal for a Qdrant WAL error (see below).

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### Ollama version mismatch warning — "client version is X.X.X"

After a Homebrew update, Ollama's CLI binary may update while the background service still runs the old version. You'll see:

```
ollama version is 0.18.0
Warning: client version is 0.22.0
```

Fix:

```
brew services restart ollama
ais_stop
ais_start
```

---

### Qdrant WAL lock — "Can't init WAL: Resource temporarily unavailable"

**What it is:** A corpus collection's write-ahead log was left locked from an unclean shutdown — force-quit terminal, power loss, or crash mid-write. Qdrant panics on the affected collection at startup. Other collections are unaffected.

**How you'll know:** Terminal shows the WAL error on `ais_start`, naming the specific collection. UI shows "Error loading corpora." Ingestion immediately fails.

## Fix:

```
ais_stop
# Delete only the collection named in the panic message — e.g. aistudio_help
rm -rf ~/qdrant_storage/collections/aistudio_help
ais_start
```

Then re-ingest the corpus via the UI (Add button).

⚠ Delete only the specific collection named in the error — not the entire `~/qdrant_storage/` folder. That would destroy all your corpora.

**Prevention:** Always stop with `ais_stop`. Never force-quit or close the terminal while AIStudio is running.

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### *Ingestion summary shows wrong file or chunk count*

Delete the corpus via the UI and re-ingest. The Qdrant collection may be out of sync with the uploads folder.

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**How do I rename a corpus?** Use the **Rename** button in the corpus header in the UI. AIStudio renames the directory, updates the corpus metadata, and triggers a background re-index. You'll see a progress estimate for large corpora. Do not rename corpus folders manually on disk — use the UI only.

**How do I rename a file inside a corpus?** AIStudio records file names at ingest time. Renaming a file after ingestion breaks citation lookup — queries will still return results but the "Open ↗" link will fail to resolve.

**Safe approach:** delete the file from the corpus using the **Delete** button in the Inspect view, then re-upload it with the correct filename via **Add**. AIStudio will re-ingest only the changed file.

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For architecture context, see [docs/architecture\\_decisions.pdf](#). For getting started, see [QUICKSTART.md](#).